DIGITAL EGYPT PIONEERS INITIATIVE (DEPI)

UK RAILWAY DATA ANALYTICS

Project Planning & Requirements Document

Dataset: UK Railway Dataset

Project Focus	Railway punctuality, delays, ticket sales, refunds, revenue, and station/route performance.
Primary Tools	Python for data cleaning and preprocessing; Power BI for data modeling, measures, and interactive reporting.
Team Leader	Banan Magdy Mohamed — Team coordination and data cleaning pipeline.
Team Members	Hossam & Elham — Data Analysis; Asmaa & Abdelrahman — Visualization and Power BI development.
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This document defines the project scope, delivery plan, team responsibilities, and 40 data-driven business requirements. Each requirement should be mapped to a validated field, measure, or report element before implementation.

1. Project Overview

The project analyzes the UK Railway Dataset using Python for data cleaning and preprocessing, followed by an interactive Power BI dashboard. The report is intended to help stakeholders understand punctuality, delay patterns, route and station performance, ticket purchasing behavior, refunds, and revenue. Findings will be descriptive and evidence-based; the project will not claim causes or impacts that the available data cannot support.

2. Project Objectives

• Profile, clean, and validate railway transaction and journey records before analysis.

• Create consistent definitions for punctuality, delay, refund, ticket-sales, and revenue KPIs.

• Build an interactive Power BI report with clear navigation, filters, and drill-down analysis.

• Answer at least 40 business questions using fields available in the supplied dataset.

• Document assumptions, data limitations, transformation steps, and calculation logic.

• Translate supported patterns into practical operational questions and recommendations.

3. Dataset Summary & Data Availability

The original proposal describes railway journey and transaction data related to delays, ticket sales, refunds, revenue, stations/routes, ticket class, purchase type, and payment method. Before implementation, the team must verify exact column names, granularity, date coverage, and whether each attribute is present in the supplied file. A question marked “Validate” below must not be presented as answerable until this check is complete.

Data area	Fields / attributes to verify	Potential analysis
Journey & time	Journey date/time; departure and arrival station/route	Traffic patterns, peak periods, route comparisons
Punctuality & delays	Delay duration/status; delay reason or cause	On-time vs delayed journeys, delay distribution and reason breakdown
Ticket sales	Ticket ID/order ID; ticket class; purchase type; payment method	Ticket mix, sales channels, payment mix
Refunds	Refund request/status; refund amount, if supplied	Refund rate and refund value patterns
Revenue	Ticket price / revenue amount; currency and refund treatment	Revenue by route, station, date, class, and channel
Reference data	Station names/IDs; route names/IDs; train/service identifiers, if present	Ranking, comparisons, and drill-through

4. Scope & Key Deliverables

• Cleaned dataset and reproducible Python preprocessing steps, with a short cleaning log.

• Documented field dictionary, data-grain decision, and data-quality checks.

• Power BI data model and reusable measures for the KPIs supported by the dataset.

• Interactive dashboard with summary, punctuality, route/station, ticket sales, and refund/revenue views where supported.

• A requirements checklist mapping all 40 questions to a visual/measure and a data-availability status.

• Executive summary with evidence-based findings, limitations, and recommendations.

5. Project Stages & Tasks

Stage	Main tasks	Owner(s)	Output
1. Planning & requirements	Confirm objectives, data grain, KPIs, 40 questions, and dashboard scope.	Whole team	Approved requirements document
2. Data profiling & cleaning	Inspect types, missing values, duplicates, invalid dates, and numeric fields; record decisions.	Banan Magdy Mohamed	Clean dataset and cleaning log
3. Exploratory analysis	Calculate distributions, trends, comparisons, and outliers; validate findings against source data.	Hossam & Elham	EDA notebook / analysis notes
4. Modeling & measures	Define relationships, date logic, and reusable measures; validate totals and denominators.	Whole team, coordinated by team leader	Model and measure definitions
5. Visualization	Create report pages, KPI cards, charts, slicers, drill-through, and consistent visual styling.	Asmaa & Abdelrahman	Interactive Power BI report
6. Testing & final delivery	Test filters, measures, labels, navigation, and consistency; prepare presentation and summary.	Whole team	Validated PBIX and final report

6. Team Roles & Responsibilities

Member(s)	Role	Responsibilities
Banan Magdy Mohamed	Team Leader & Data Cleaning Specialist	Coordinate work and milestones; lead cleaning/preprocessing; handle missing values, date formatting, duplicate checks, and data integrity documentation.
Hossam & Elham	Data Analysts	Perform EDA and supported statistical analysis; identify trends, correlations, and performance patterns; map findings to requirements.
Asmaa & Abdelrahman	Visualization & Power BI Developers	Design report layout and visuals; implement filters/interactions; build and test report pages; ensure clear data storytelling.

7. Tools & Technologies

Tool	Purpose
Python	Data profiling, cleaning, preprocessing, exploratory analysis, and validation.
Pandas / NumPy (as needed)	Tabular transformations, missing-value checks, duplicate checks, and calculations.
Power BI Desktop	Data modeling, DAX measures, interactive report design, and final dashboard.
Power Query / DAX	Repeatable transformations and KPI calculations within Power BI.
GitHub	Version control and documentation for code, reports, and project materials.

8. Business Questions / Requirements (40)

The following 40 requirements form the minimum analysis checklist. “Validate” means the field or calculation must be confirmed against the actual supplied dataset before the requirement is marked complete. If a field is absent, record the requirement as a data limitation rather than inventing a value.

ID	Business question / requirement	Suggested output	Data status / validation
01	What is the total recorded ticket revenue for the selected period?	KPI card	Validate revenue/price field
02	How does ticket revenue change over time (day, week, or month)?	Line chart	Validate date and revenue fields
03	Which month has the highest recorded ticket revenue?	Monthly trend / KPI	Validate date and revenue fields
04	Which month has the lowest recorded ticket revenue?	Monthly trend / KPI	Validate date and revenue fields
05	How many distinct journeys or transactions are recorded?	KPI card	Validate journey/transaction ID and data grain

ID	Business question / requirement	Suggested output	Data status / validation
06	What is the average ticket value?	KPI card	Validate ticket value and ticket/transaction ID
07	How does the number of journeys vary by date?	Line chart	Validate journey date and unique journey key
08	Which days or time periods have the highest journey volume?	Column chart / heatmap	Validate date/time fields
09	What percentage of journeys are delayed versus on time?	KPI cards / donut	Validate delay status or delay duration
10	What is the average delay duration?	KPI card	Validate delay duration field
11	How is delay duration distributed across journeys?	Histogram / column chart	Validate delay duration field
12	Which delay reasons are recorded most frequently?	Ranked bar chart	Validate delay-reason field
13	Which recorded delay reason is associated with the greatest total delay duration?	Ranked bar chart	Validate reason and duration fields
14	How does the delay rate change over time?	Line chart	Validate delay status and date
15	Which routes have the highest number of delayed journeys?	Ranked bar chart	Validate route and delay fields
16	Which routes have the lowest delay rate?	Ranked bar chart	Validate route and journey count
17	How does punctuality compare across departure stations?	Bar chart / matrix	Validate station and punctuality fields
18	How does punctuality compare across arrival stations?	Bar chart / matrix	Validate station and punctuality fields
19	Which stations have the highest journey volume?	Ranked bar chart	Validate station and journey key
20	Which routes have the highest journey volume?	Ranked bar chart	Validate route and journey key
21	How does revenue compare across routes?	Ranked bar chart	Validate route and revenue fields
22	How does revenue compare across departure stations?	Bar chart	Validate station and revenue fields
23	How does revenue compare across arrival stations?	Bar chart	Validate station and revenue fields
24	What share of total revenue comes from each route?	100% stacked chart / table	Validate route and revenue fields
25	How are ticket sales distributed by ticket class?	Bar / donut chart	Validate ticket-class field
26	Which ticket class contributes the most recorded revenue?	Ranked bar chart	Validate ticket class and revenue
27	How does delay rate differ by ticket class?	Clustered bar chart	Validate ticket class and delay fields
28	How are tickets distributed by purchase type (for example, online vs station counter)?	Bar / donut chart	Validate purchase-type field
29	Which purchase type generates the highest recorded revenue?	Ranked bar chart	Validate purchase type and revenue
30	How does refund-request frequency vary by purchase type?	Bar chart	Validate refund and purchase-type fields
31	How are tickets distributed by payment method?	Bar / donut chart	Validate payment-method field
32	Which payment method accounts for the highest recorded revenue?	Ranked bar chart	Validate payment method and revenue
33	How many refund requests are recorded?	KPI card	Validate refund-request/status field
34	What percentage of recorded tickets or transactions have a refund request?	KPI card	Validate refund field and denominator
35	How does refund-request frequency change over time?	Line chart	Validate refund and date fields

ID	Business question / requirement	Suggested output	Data status / validation
36	Which routes have the highest number of refund requests?	Ranked bar chart	Validate route and refund fields
37	How does refund-request rate differ between delayed and on-time journeys?	Clustered bar chart	Validate delay and refund fields; association only
38	What is the recorded refund value by month, route, or ticket class?	Line / bar chart	Validate refund amount field; otherwise unavailable
39	Can users filter the report by date, route, station, ticket class, purchase type, and payment method where available?	Slicers / filter panel	Required report functionality; verify each field
40	Do KPI totals and percentages remain consistent across filters and match independent source-data checks?	Validation checklist	Required quality control

9. Proposed Dashboard Pages

Page	Purpose	Suggested components
1. Executive Overview	Summarize service and commercial performance.	Total revenue, journey count, delay rate, average delay, refund-request count; trend charts and high-level route/station comparisons.
2. Punctuality & Delays	Explore the scale, timing, and recorded reasons for delays.	On-time vs delayed split, delay-duration distribution, delay trend, reason breakdown, route/station comparisons.
3. Route & Station Performance	Compare volume, punctuality, and revenue across locations.	Top routes/stations, departure vs arrival station analysis, route-level performance matrix, drill-through where supported.
4. Ticket Sales & Customer Choices	Understand ticket mix and transaction patterns.	Ticket class, purchase type, payment method, revenue contribution, date trends.
5. Refunds & Revenue	Review recorded refund patterns alongside revenue.	Refund-request KPI/rate, refund trends, route/class/channel breakdowns; refund value only if present.
6. Details & Validation (optional)	Support investigation and transparent QA.	Detailed records, filter controls, KPI definitions, data limitations, and validation notes.

10. Proposed KPI Definitions

KPI	Suggested definition	Important note
Total Recorded Revenue	Sum of the validated ticket price/revenue field.	Confirm currency, granularity, and whether refunds are already deducted.
Journey / Transaction Count	Distinct count of the appropriate journey or transaction identifier.	Do not count rows as journeys if each journey can contain multiple line items.
Delay Rate	Delayed journeys divided by all eligible journeys in the same filter context.	Define “delayed” from the dataset's status or agreed threshold; document the rule.
Average Delay Duration	Average delay duration for eligible records with a valid duration.	Check units, missing values, negative values, and whether on-time journeys are represented as zero.
On-Time Rate	On-time journeys divided by all eligible journeys in the same filter context.	Use a consistent denominator and exclude invalid records only with documented rationale.
Refund Request Count	Distinct count of refund requests or affected transactions, based on available identifiers.	Do not treat repeated refund rows as separate requests unless the data grain supports it.
Refund Request Rate	Transactions with a refund request divided by eligible transactions.	State whether the denominator is tickets, journeys, or transactions.
Revenue Contribution %	Revenue for the selected group divided by total revenue in the current comparison context.	Document how slicers affect the denominator.

11. Data Quality Checks & Assumptions

• Confirm the actual file schema, row count, date coverage, and record granularity before analysis.

• Parse journey and transaction dates/times consistently; inspect ambiguous formats and invalid timestamps.

• Check missing values in identifiers, station/route fields, delay fields, ticket attributes, revenue, and refund fields.

• Check duplicates using the correct grain. Repeated journey or order identifiers may be valid when records represent multiple tickets or line items.

• Ensure revenue, price, and delay-duration fields are numeric and use consistent units and currency.

• Investigate negative values, implausible delay durations, invalid categories, and outliers before deciding whether to correct or exclude them.

• Define the rules for delayed/on-time classification and the denominators used for every percentage.

• Do not claim that a delay reason caused a delay, or that delays caused refunds, based on descriptive association alone.

• If refund amount, ticket class, purchase type, payment method, route, or station fields are absent, mark related questions as unavailable or request additional data.

• Document all transformations and assumptions so the analysis can be reproduced.

12. Acceptance Criteria

- The cleaned dataset or preprocessing workflow runs reproducibly and has documented transformation steps.
- The data model and core measures use clearly defined identifiers, date logic, and denominators.
- All 40 requirements are documented and mapped to a visual, measure, filter, validation check, or explicit data limitation.
- The dashboard includes clear titles, readable labels, consistent formats, and appropriate visual choices.
- Available slicers and drill-downs behave consistently across relevant report pages.
- Core totals and percentages are checked against the cleaned source or independent calculations.
- Unsupported questions are clearly flagged; no unavailable metric is presented as if it were calculated.
- The team reviews and integrates each member's contribution into one final Power BI report.

13. Suggested Final Submission Package

- Cleaned dataset and/or Python preprocessing notebook/scripts.
- Data dictionary, data-quality notes, and KPI/measure definitions.
- Power BI (.pbix) file with report pages and interactive filters.
- This Project Planning & Requirements Document with the 40-question checklist.
- Short executive summary with key supported findings, limitations, and recommendations.
- Final presentation demonstrating the dashboard and explaining each team member's contribution.

Document note: This is a requirements and planning document. Final data-availability statuses and numerical findings must be confirmed against the exact railway dataset used by the team.